Department of computer science

Event Driven Programming

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  - Using the DataGrid View for database access

Assessment Methods

- Assignment/quizzes 10 %
- Mid Semester Examination 20%
- Project 20%
- Final Examination 50%

Reference

Text books and References:

1. An introduction to programming using visual basic 6.0, fourth edition, David I. Schneider Evjen, B et al, (2008). Professional Visual Basic 2008.
2. Crosspoint Boulevard: Wiley Publishing Inc. Gary Cornell and Jonathan Morrison (2002).
3. Programming VB.NET: A Guide for Experienced Programmers. USA: APress Cameron Wakefield, Henk-Evert Sonder and Wei Meng Lee. VB.NET Developers Guide. USA: Syngress Publishing, Inc

Chapter One

Introduction

Chapter Content

- Introduction
  - Introduction to Software Development
  - Software Development Approaches
  - Rapid Application Development
  - Software Development Principles

1.1 Event Driven Programming

- Event-driven programming focuses on events
- Event-driven programming is a paradigm where entities
  - (objects, services, and so on) communicate indirectly by sending messages to one another through an intermediary.
- Event:
  - A signal to the program that something has happened.

Cont..

- It can be triggered either by external user actions, such as
  - mouse movements
  - button clicks, and keystrokes,
  - by the operating system, such as a timer.
- The program can choose to respond to or ignore an event.

1.2 Introduction to Software Development

- What is the introduction of software development?
- Software development is the process of
  - conceiving, specifying, designing, programming, documenting, testing, and bug fixing involved in creating and maintaining applications, frameworks, or other software components.

Cont..

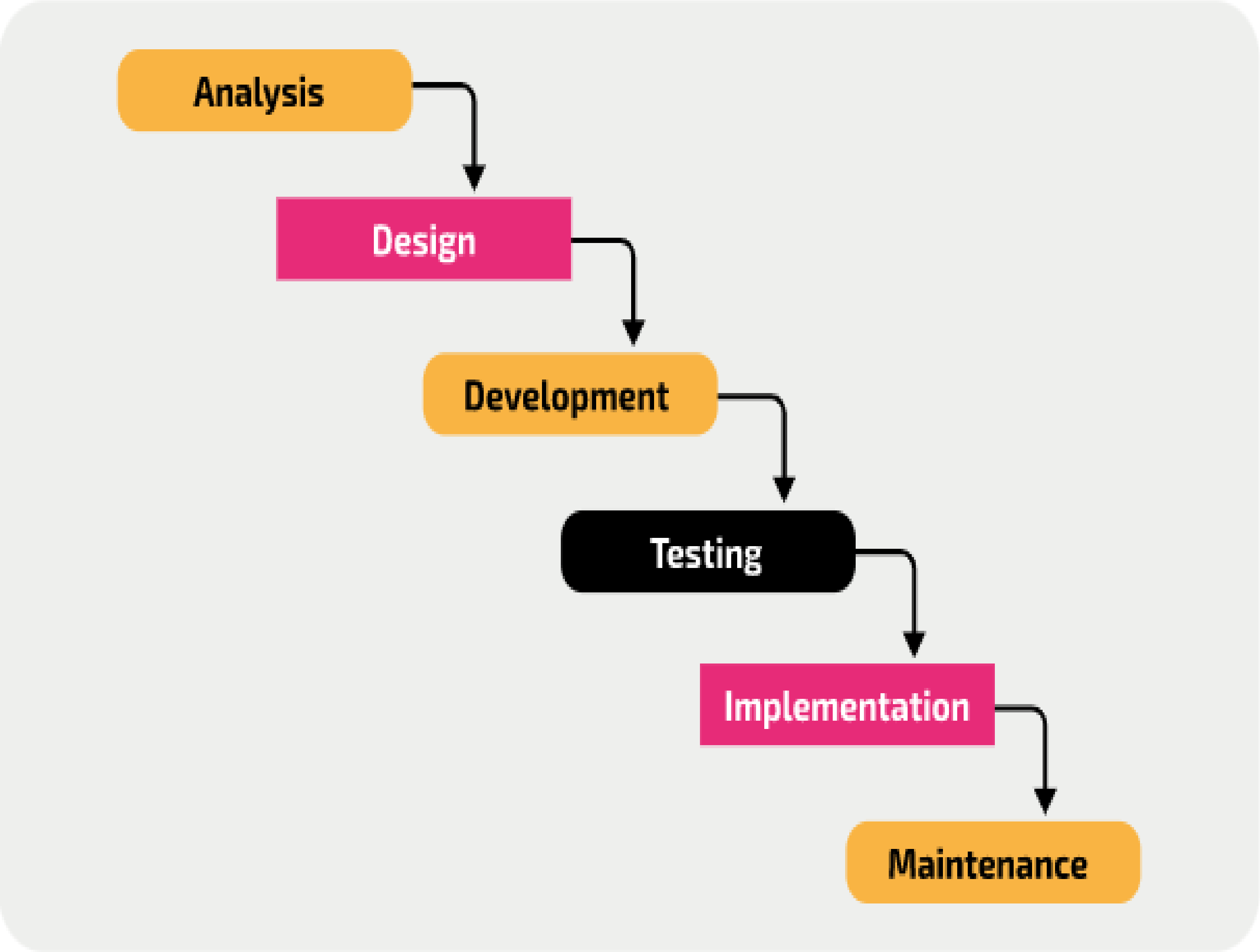
- Software development is the process programmers use to build computer programs.
- The process, also known as the
  - Software Development Life Cycle (SDLC),
  - includes several phases that provide a method for building products that meet technical specifications and user requirements.

1.2.1 Software Development Process

- The process vary in the detail of what activities they prescribe, their related artefacts.(b/c there are many approaches)
- However the most common process is
  - **Analysis**
  - **Design**
  - **Implementation**
  - **Testing**
  - **Deployment**

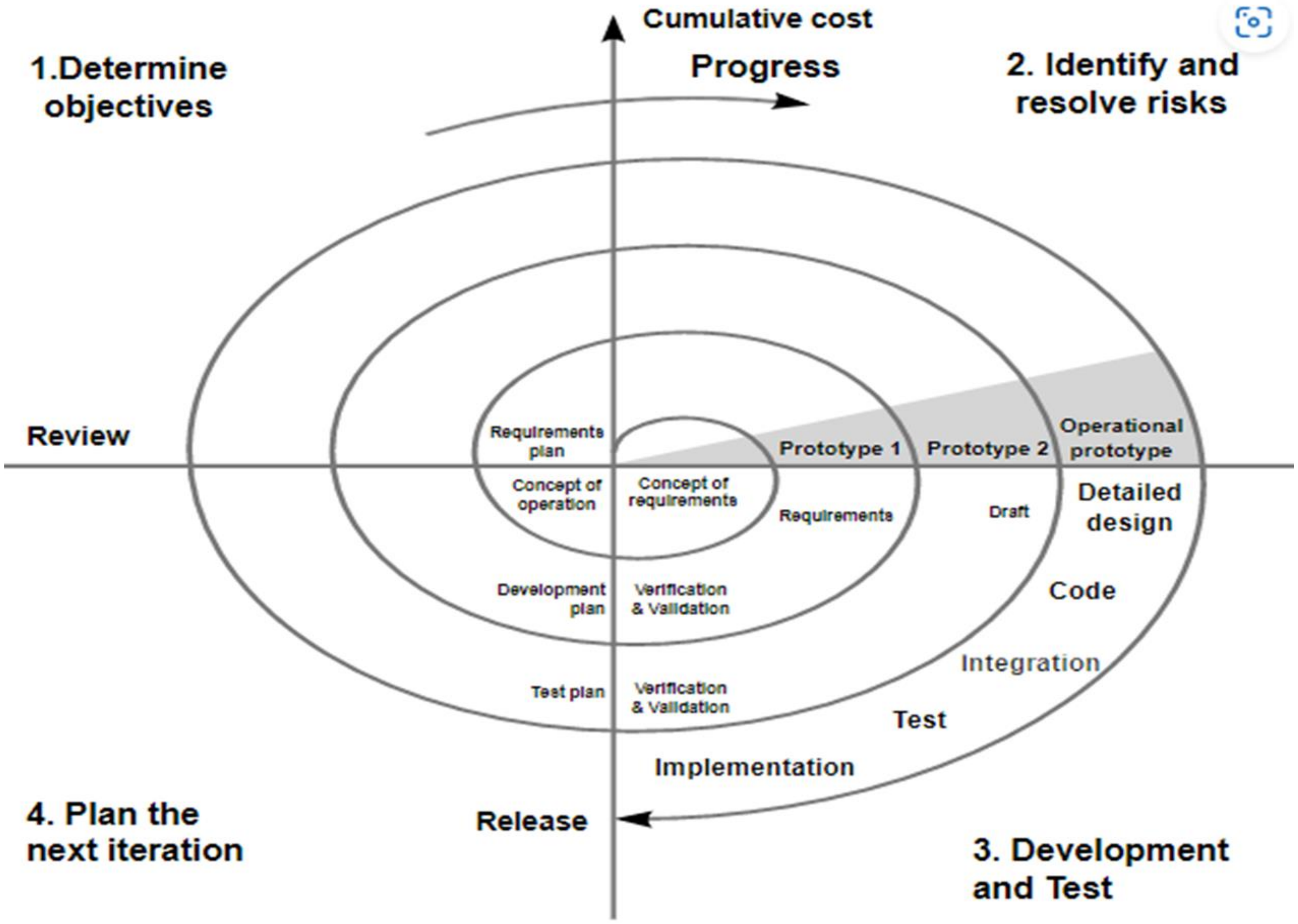
1.3 Software Development Approaches

- the most commonly used software development approaches are
  - **Waterfall Approach**
  - **Spiral Approach**
  - **RAD**
  - **Agile**
  - **Incremental**
  - **Etc…**



Waterfall Approach

- The stages of a project were carried out **sequentially**.
- The entirety of each stage had to be completed, formally documented, discussed, agreed, and signed off before the next stage could begin.
- This has been described as the **waterfall lifecycle**.



Spiral Approach

- is a development method that uses the same steps as the waterfall method, but also uses **project cycles**, each culminating in a version of the software (a prototype) that is formally reviewed to inform the next cycle.

Determine objectives

Identify and resolve risks

Develop and test

Plan (the next iteration)

Reading Assignment

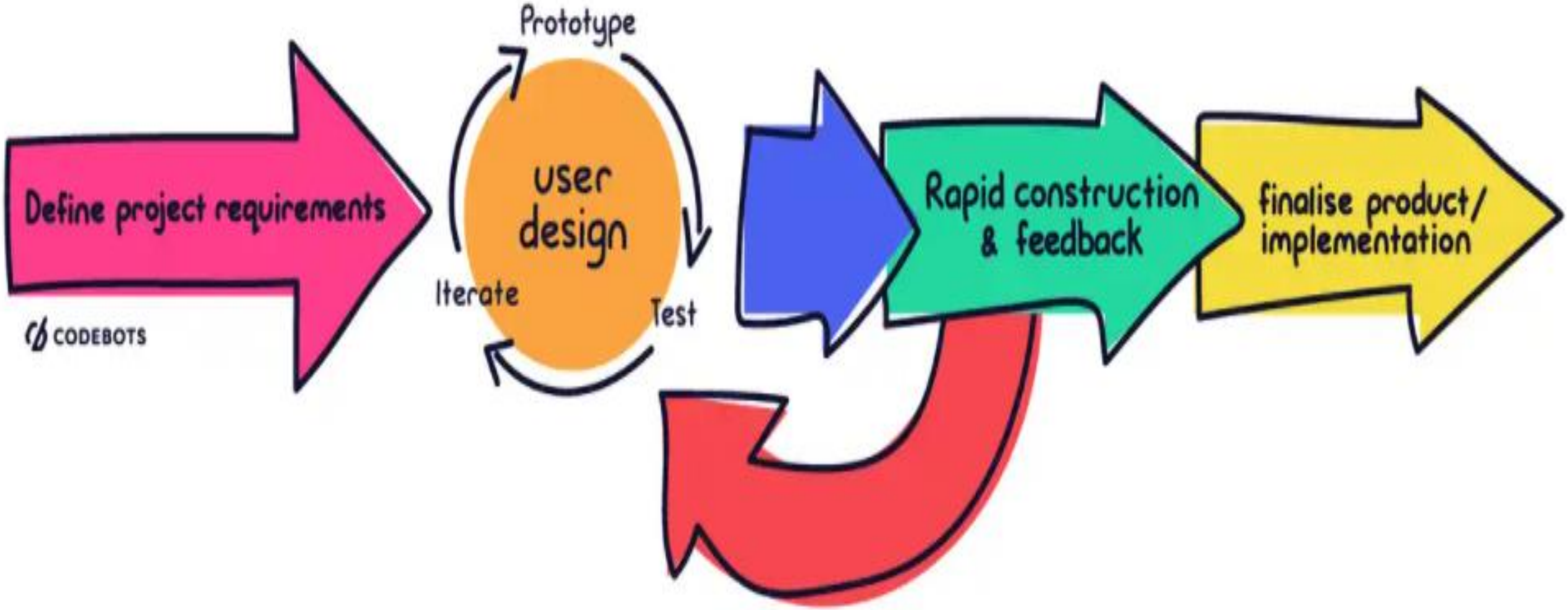
- Incremental approach
- Agile Approach
- Extreme programming
- DevOps Methodology
- Joint Application Development Methodology

1.4 Rapid application development

- is a development model prioritizes rapid prototyping and quick feedback over long drawn out development and testing cycles.
- With rapid application development, developers can make multiple iterations and updates to a software rapidly without needing to start a development schedule from scratch each time.

Steps in RAD

- Although RAD has changed over the years, these four basic steps provide some continuity over the years.
  1. **Define the requirements**
  2. **Prototype** -> *the actual development takes place.*
     - Instead of following a strict set of requirements
  3. **Receive Feedback**
     - *Feedback on what's good, what's not, what works, and what doesn't*
  4. **Finalize Software**


Define project requirements
CODEBOTS
Prototype
user design
Iterate
Test
Rapid construction & feedback
finalise product/ implementation

Cont..

- Advantages of the RAD model
  - helps to reduce the risk and required efforts on the part of the software developer.
  - helps the clients to take quick reviews for the project.
  - Encourages customer feedback which always provides improvement scope for any software development project.
  - Each phase in RAD delivers the highest priority functionality to the client.

Cont..

- Disadvantages RAD model
  - Depends on the strong team and individual performances for clearly identifying the exact requirement of the business.
  - This approach demands highly skilled developers and a designer' s team which may not be possible for every organization.
  - This is not applicable for the developer to use in small budget projects as the cost of modeling and automated code generation is very high.
  - Progress and problems accustomed are hard to track as such there is no documentation to demonstrate what has been done.

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Chapter Two

Introduction to .NET

Chapter Content

- Chapter 2: Introduction to .NET
  - The .NET Platform and Its Architecture
    - Common Language Runtime
    - Base Class Library
  - Uses of .NET Platform in Application Development
  - Introduction to Microsoft Visual Studio 2010

What is .NET Framework?

- **.Net Framework** is a *software development platform developed by Microsoft for building and running Windows applications.*
- The .Net framework consists of ***developer tools, programming languages, and libraries to build desktop and web applications.***
- It is also used to build websites, web services, and games.
- The .Net framework was meant to *create applications, which would run on the Windows Platform.*
- The first version of the .*Net framework was released in the year 2002*.
- The version was called .Net framework 1.0.

Cont.…

- .NET Framework Architecture is a programming model for the .Net platform that provides an execution environment and integration
- Has various programming languages for simple development and deployment of various Windows and desktop applications.
- It consists of class libraries and reusable components.

.NET Components

- Common Language Runtime
  - **CLR** is the execution engine that handles running applications.
  - It provides services like thread management, garbage collection, type-safety, exception handling, and more.
- also knows as the runtime.
- CLR’ s responsibility is to mange any code written in .NET languages such as C#, VB.NET, or F#.

Base Class Library(BCL)

- The Class Library provides a set of APIs and types for common functionality.
- It provides types for strings, dates, numbers, etc.
- The Class Library includes APIs for reading and writing files, connecting to databases, drawing, and more.
- The .NET Framework includes a set of standard class libraries.
- A class library is a collection of methods and functions that can be used for the core purpose.

Characteristics of .NET Frame work

Uses of .NET Platform in Application Development

- Can be used to develop
  - Windows applications (windows forms)
  - Web (Asp.net)
  - Mobile applications that run .NET
  - Ado.net For Data base connectivity

.NET Programming languages?

- Supports he following Languages
  - Visual Basic
  - C#
  - F#
  - Visual C++
  - Etc….

1.5 Visual Basic programming language

- **VB.NET** is also known as **Visual Basic.NET**.
  - It stands for *Visual Basic .Network Enabled Technologies*.
- ***VB.NET*** is a simple, multi-paradigm object-oriented programming language designed to create a wide range of
  - Windows,
  - Web,
  - Mobile applications built on the **.NET Framework**.
- *developed by Microsoft in 2002.*

VB.NET Features

- It is an object-oriented programming language
- This language is used to design user interfaces for window, mobile, and web-based applications.
- It supports a *rapid application development tool kit.*
- It is not a case sensitive language like other languages such as C++, java, etc.
- It provides simple events management in .NET application.
- A Window Form enables us to inherit all existing functionality of form that can be used to create a new form. So, in this way, it reduced the code complexity.

Advantages of VB.NET

- Your code will be formatted automatically.
- You will use object-oriented constructs to create an enterprise-class code.
- You can create web applications *with modern features like performance counters, event logs, and file system.*
- You can connect your applications to *other applications created in languages that run on the .NET framework.*
- You will enjoy features like *docking, automatic control anchoring, and in-place menu editor all good for developing web applications.*

```
Dim x As Double
Dim Y As Double
Dim ans As Double

Sub Add()
    Console.WriteLine(" please enter the first number ")
    x = Console.ReadLine()
    Console.WriteLine(" please enter the second number ")
    Y = Console.ReadLine()
    ans = x + Y
    Console.WriteLine(" The submision is  " & ans)
End Sub
Sub SUBS()
    Console.WriteLine(" please enter the first number ")
    x = Console.ReadLine()
    Console.WriteLine(" please enter the second number ")
    Y = Console.ReadLine()
    ans = x - Y
    Console.WriteLine(" The submision is  " & ans)
End Sub
Sub MUL()
    Console.WriteLine(" please enter the first number ")
    x = Console.ReadLine()
    Console.WriteLine(" please enter the second number ")
    Y = Console.ReadLine()
    ans = x * Y
    Console.WriteLine(" The submision is  " & ans)
End Sub
Sub DIV() ...
```

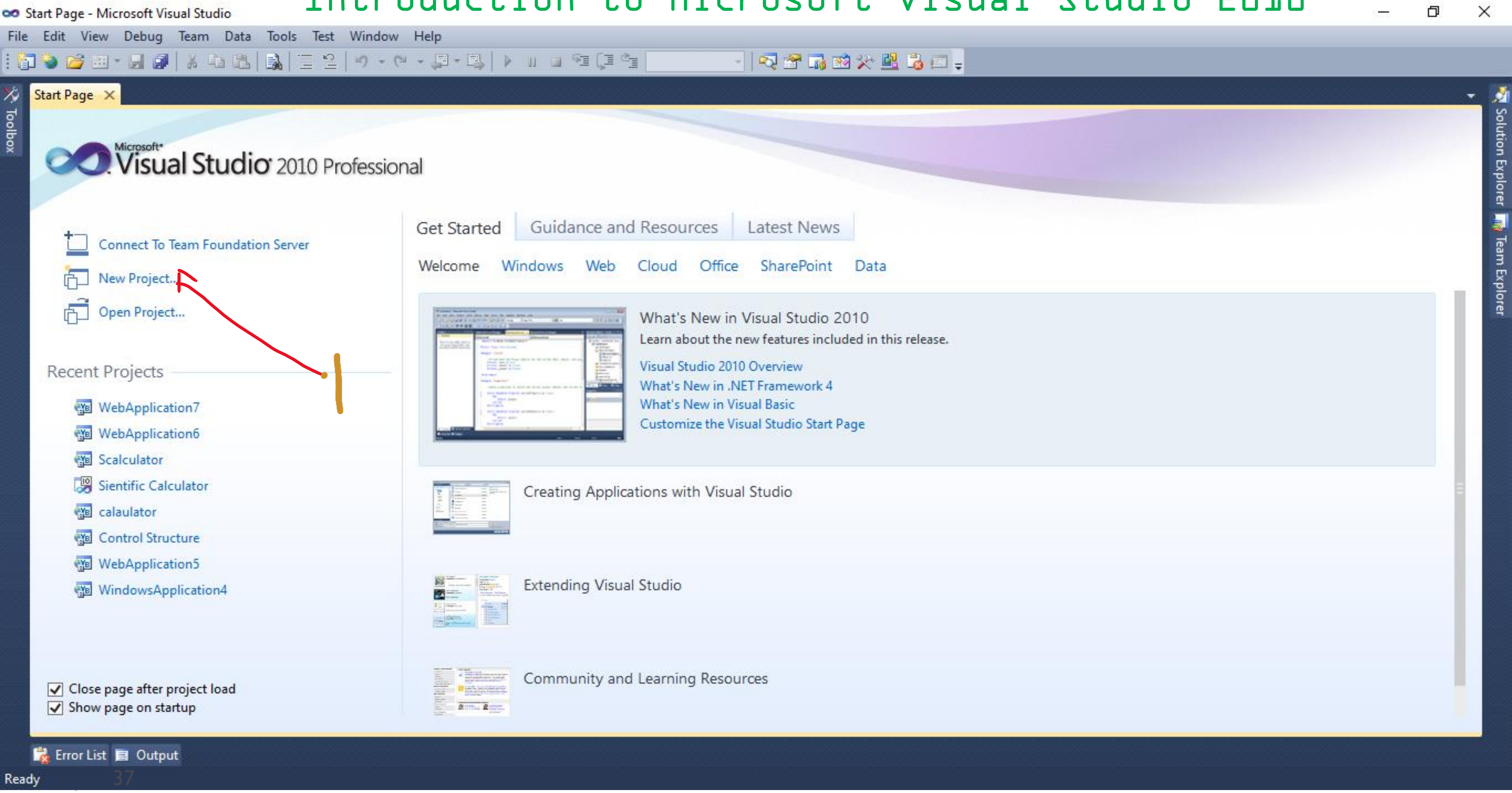
Disadvantages of VB.NET

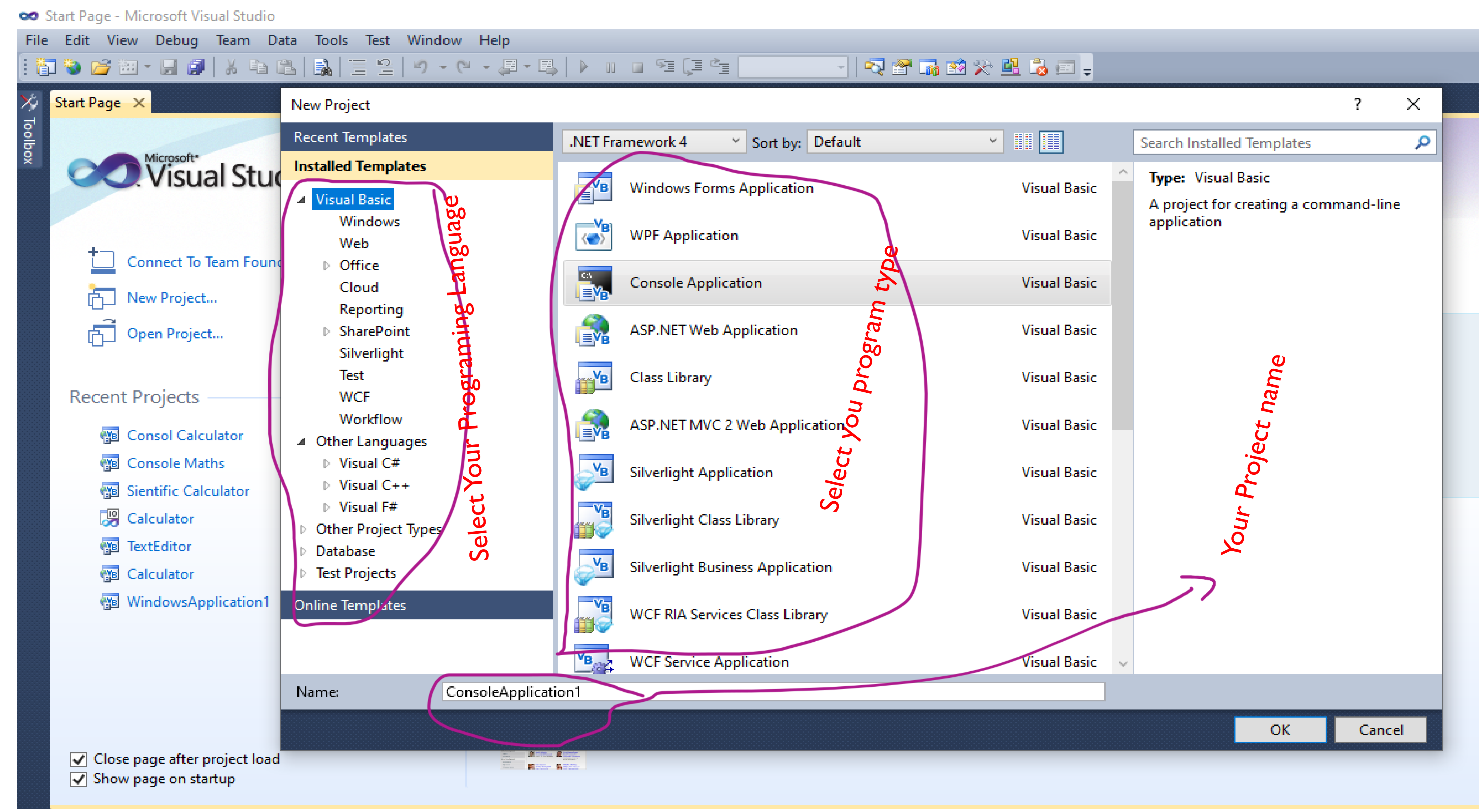
- VB.NET cannot handle pointers directly.
- This is a significant disadvantage since pointers are much necessary for programming.
- Any additional coding will lead to many CPU cycles, requiring more processing time.
- Your application will become slow.

Summary:

- VB.NET was developed by Microsoft.
- It is an object-oriented language.
- The language *is not case sensitive*.
- VB.NET programs run on the .NET framework.
- In VB.NET, the ***garbage collection process has been automated.***
- The language provides windows forms from which you can inherit your own forms.
- VB.NET allows you to enjoy the drag and drop feature when creating a user interface.

Introduction to Microsoft Visual Studio 2010




Start Page - Microsoft Visual Studio
File
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New Project...
Open Project...
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Consol Calculator
Console Maths
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TextEditor
Calculator
WindowsApplication1
Close page after project load
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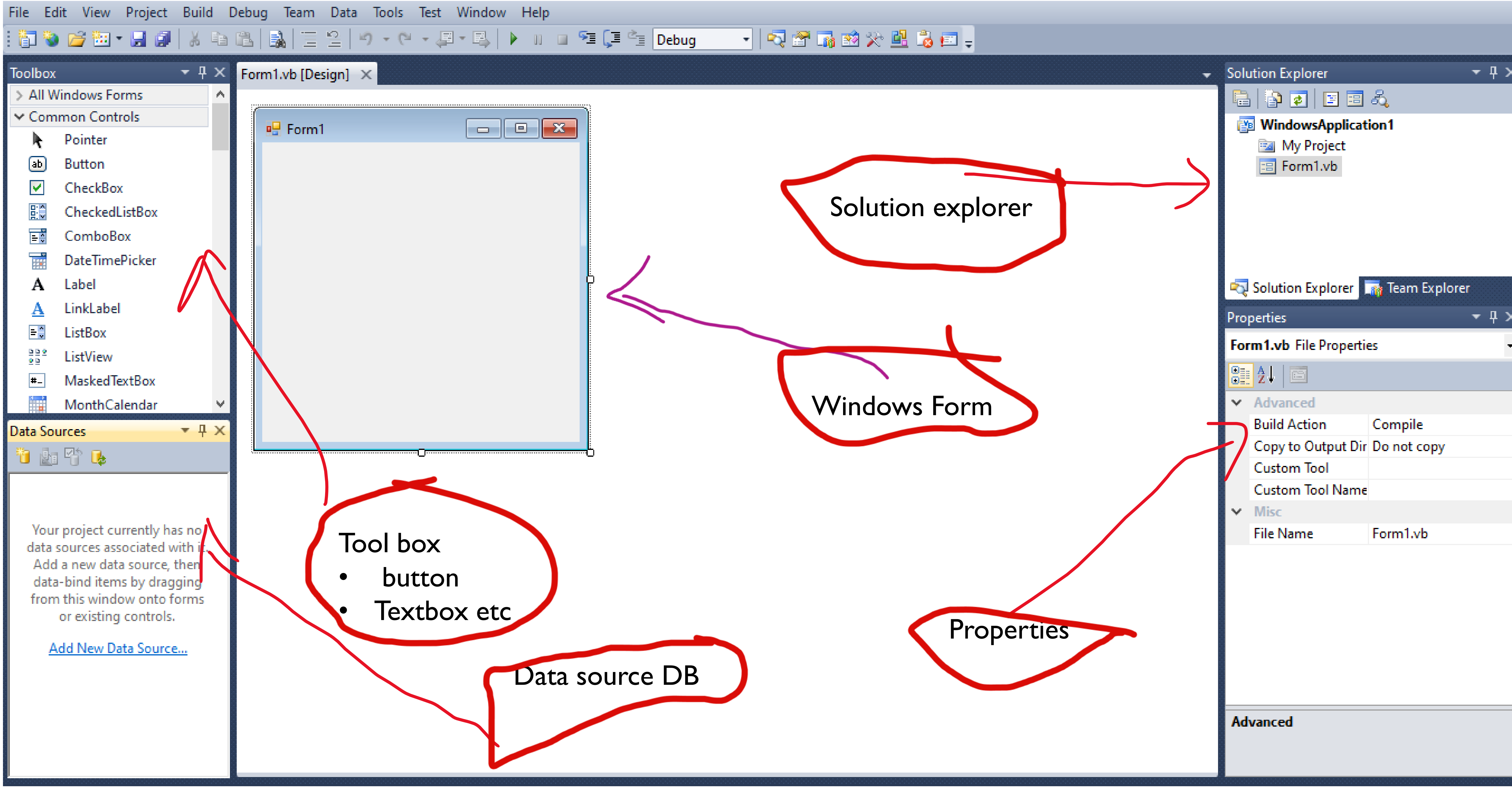
New Project
?
Recent Templates
Installed Templates
Visual Basic
Windows
Web
Office
Cloud
Reporting
SharePoint
Silverlight
Test
WCF
Workflow
Other Languages
Visual C#
Visual C++
Visual F#
Other Project Types
Database
Test Projects
Online Templates
.NET Framework 4
Sort by:
Default
Search Installed Templates
Windows Forms Application
Visual Basic
WPF Application
Visual Basic
Console Application
Visual Basic
ASP.NET Web Application
Visual Basic
Class Library
Visual Basic
ASP.NET MVC 2 Web Application
Visual Basic
Silverlight Application
Visual Basic
Silverlight Class Library
Visual Basic
Silverlight Business Application
Visual Basic
WCF RIA Services Class Library
Visual Basic
WCF Service Application
Visual Basic
Type: Visual Basic
A project for creating a command-line application
Name:
ConsoleApplication1
OK
Cancel
Select Your Programing Language
Select you program type
Your Project name


File Edit View Project Build Debug Team Data Tools Test Window Help
Debug
Toolbox
All Windows Forms
Common Controls
Pointer
Button
CheckBox
CheckedListBox
ComboBox
DateTimePicker
Label
LinkLabel
ListBox
ListView
MaskedTextBox
MonthCalendar
Data Sources
Your project currently has no data sources associated with it. Add a new data source, then data-bind items by dragging from this window onto forms or existing controls.
Add New Data Source...
Form1.vb [Design]
Form1
Solution Explorer
WindowsApplication1
My Project
Form1.vb
Solution Explorer
Team Explorer
Properties
Form1.vb File Properties
Advanced
Build Action
Compile
Copy to Output Dir
Do not copy
Custom Tool
Custom Tool Name
Misc
File Name
Form1.vb
Advanced
Solution explorer
Windows Form
Tool box
• button
• Textbox etc
Data source DB
Properties

SDI and MDI Forms

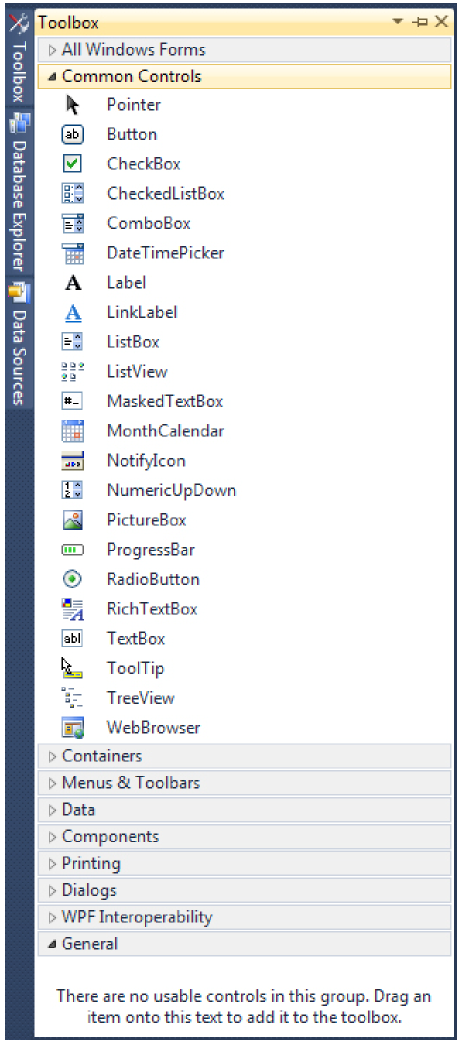
- Any windows based environment supports two types of application.
  - SDI Application (Single Document Interface Application)
  - MDI Application (Multiple Document Interface Application)
- **SDI Applications:** The applications that will allow the user to work with only one file / document at once are known as SDI Applications.
  - **Ex:** Notepad, Paintbrush, Wordpad etc.
- **MDI Applications:** The applications that will allow the user to work with more than one file / Document at once are known as MDI Applications.
  - **Ex:** MS- Word, MS – Excel, MS - Powerpoint etc.
- To create any MDI App., we need to include MDI form with in the application.
- MDI form will be treated as parent form.
- Remaining forms will be treated as child forms.

Add controls to a Blank window

The first thing we'll do is to add a button to the blank form. We'll then write a single line of code, so that you can see how things work.

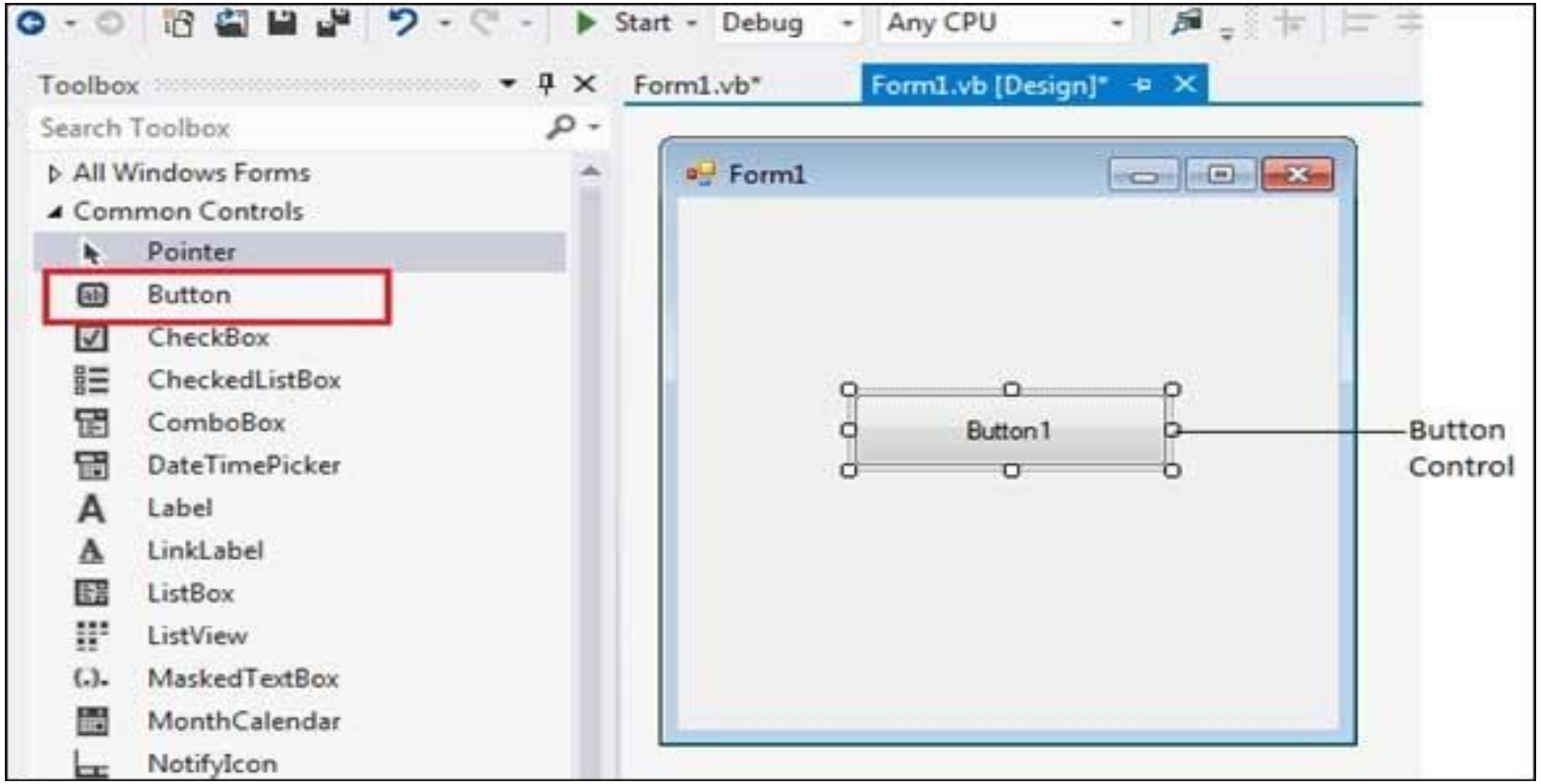
Tool box

- The tool box contains all the necessary controls
- If you want to add a control to a form, you can use the Toolbox on the left of Visual Studio. Move your mouse over to the Toolbox, and click the arrow symbol (or plus symbol) next to **Common Controls. You should see the following list of things that you can add to your form:**


Toolbox
Toolbox
Database Explorer
Data Sources
All Windows Forms
Common Controls
Pointer
Button
CheckBox
CheckedListBox
ComboBox
DateTimePicker
Label
LinkLabel
ListBox
ListView
MaskedTextBox
MonthCalendar
NotifyIcon
NumericUpDown
PictureBox
ProgressBar
RadioButton
RichTextBox
TextBox
ToolTip
TreeView
WebBrowser
Containers
Menus & Toolbars
Data
Components
Printing
Dialogs
WPF Interoperability
General
There are no usable controls in this group. Drag an item onto this text to add it to the toolbox.

Button Control

- The button control represents a standard windows button .
- It is generally used to generate a click events by providing a handler for the click event



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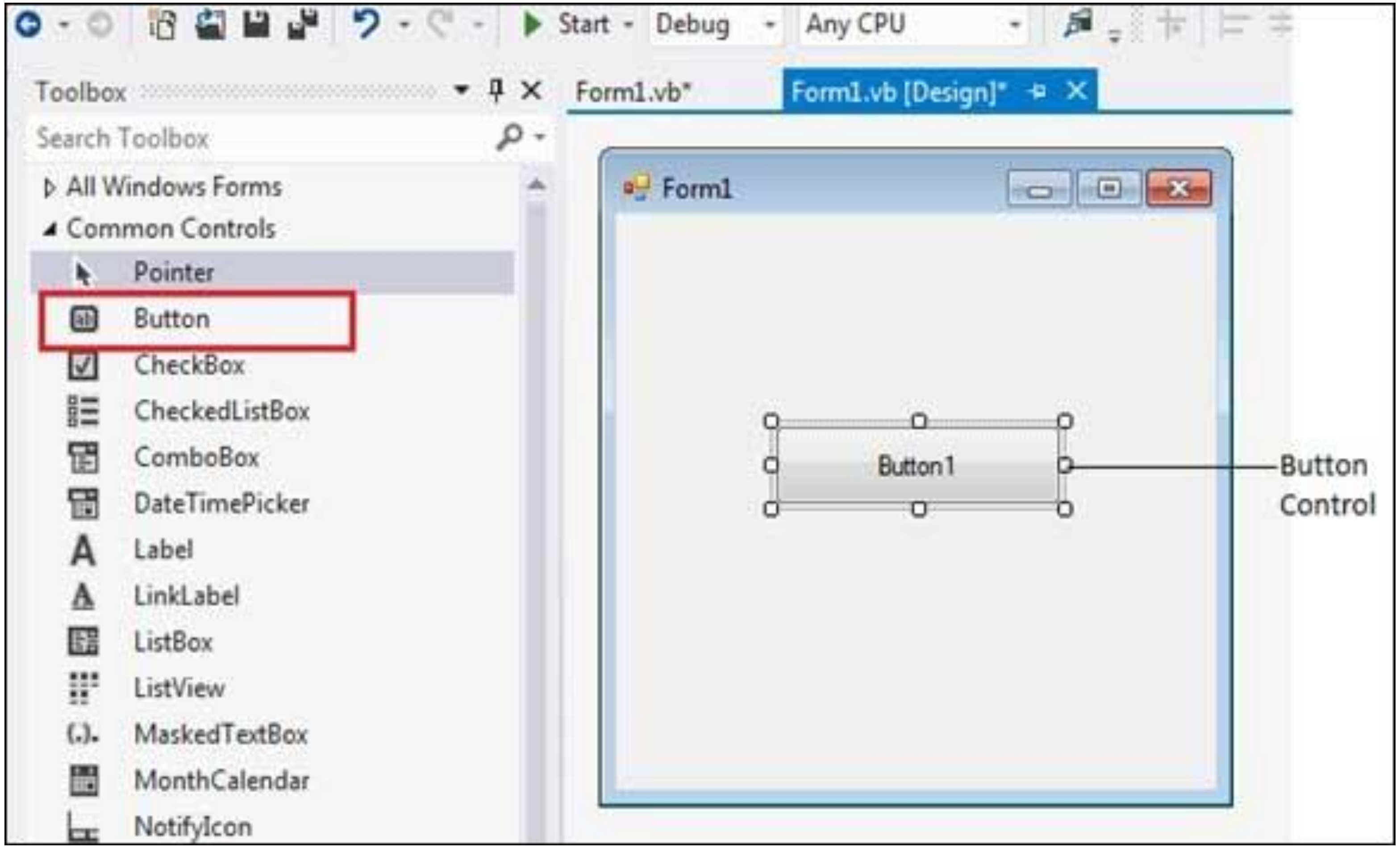


Start
Debug
Any CPU
Toolbox
Form1.vb*
Form1.vb [Design]*
Search Toolbox
All Windows Forms
Common Controls
Pointer
Button
CheckBox
CheckedListBox
ComboBox
DateTimePicker
Label
LinkLabel
ListBox
ListView
MaskedTextBox
MonthCalendar
NotifyIcon
Form1
Button1
Button Control

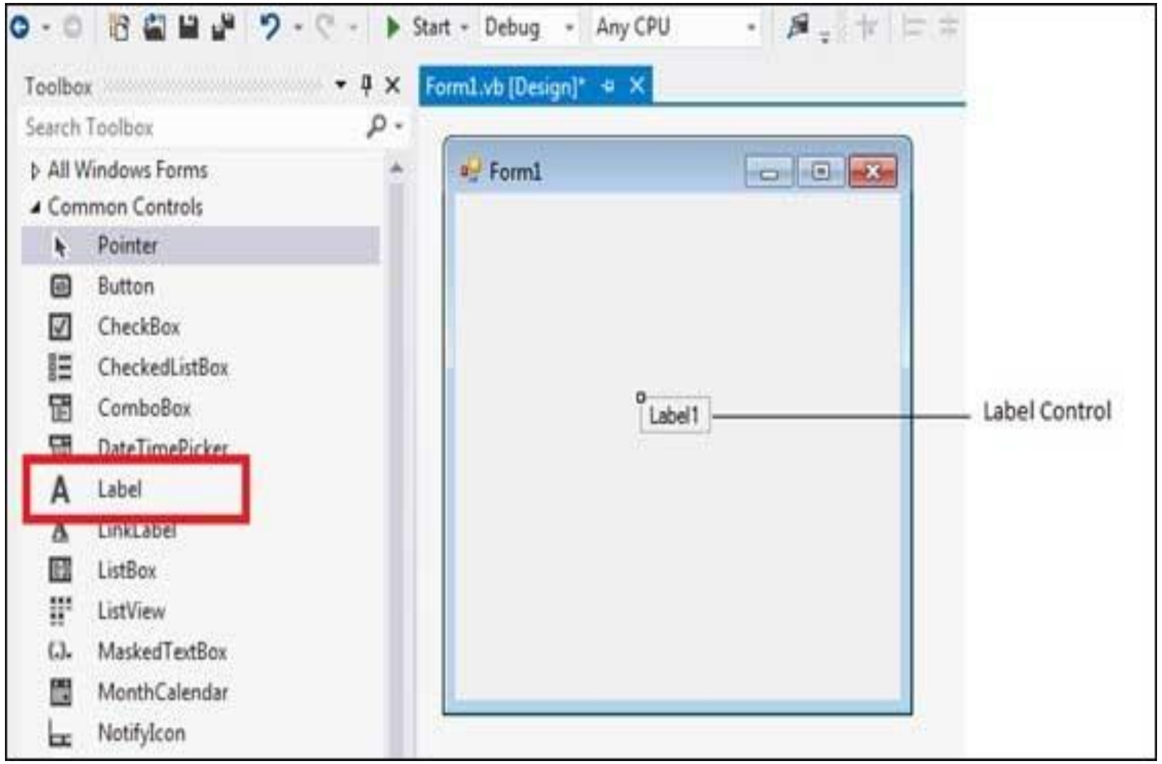
Event associated with Button

Event

- An event is an member of a class, used to perform required action
- Event is similar to function but pointed by a delegate or function pointer
- Every event will have particular period of time that is to be called as event raising/firing
- Delegate are completely backbone for the event

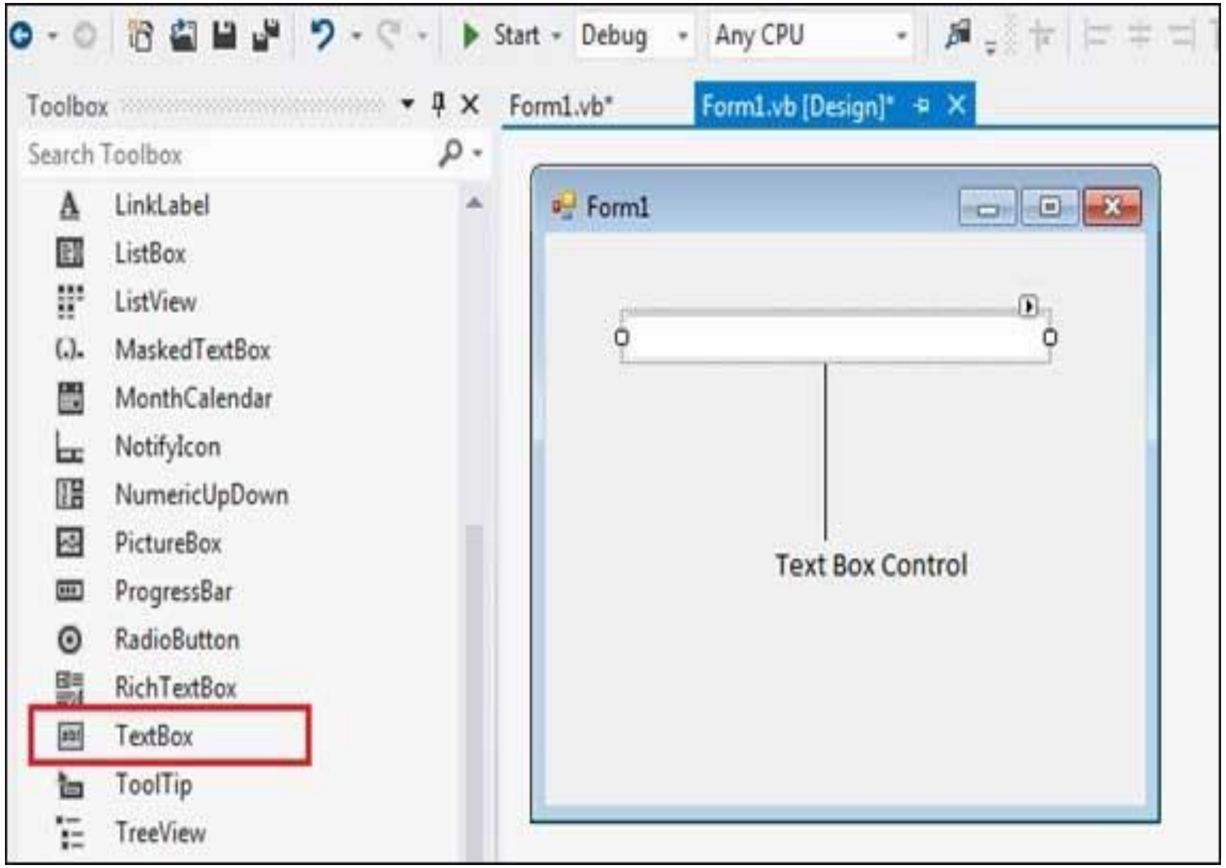
Label Control

- The label control is one of the most commonly used windows form
- It is generated used to display the text that is not supposed to be changed by the user at run time



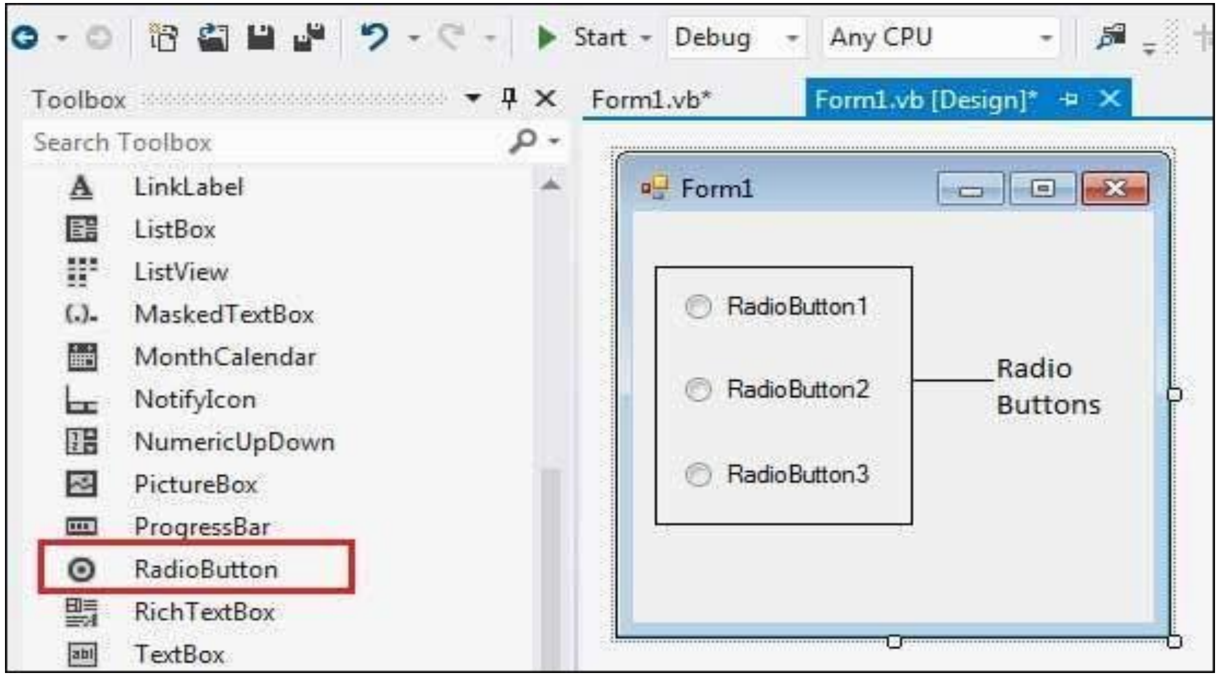
TextBox Control

- This control is used to accept the data from the user and also to display the data dynamically to the user



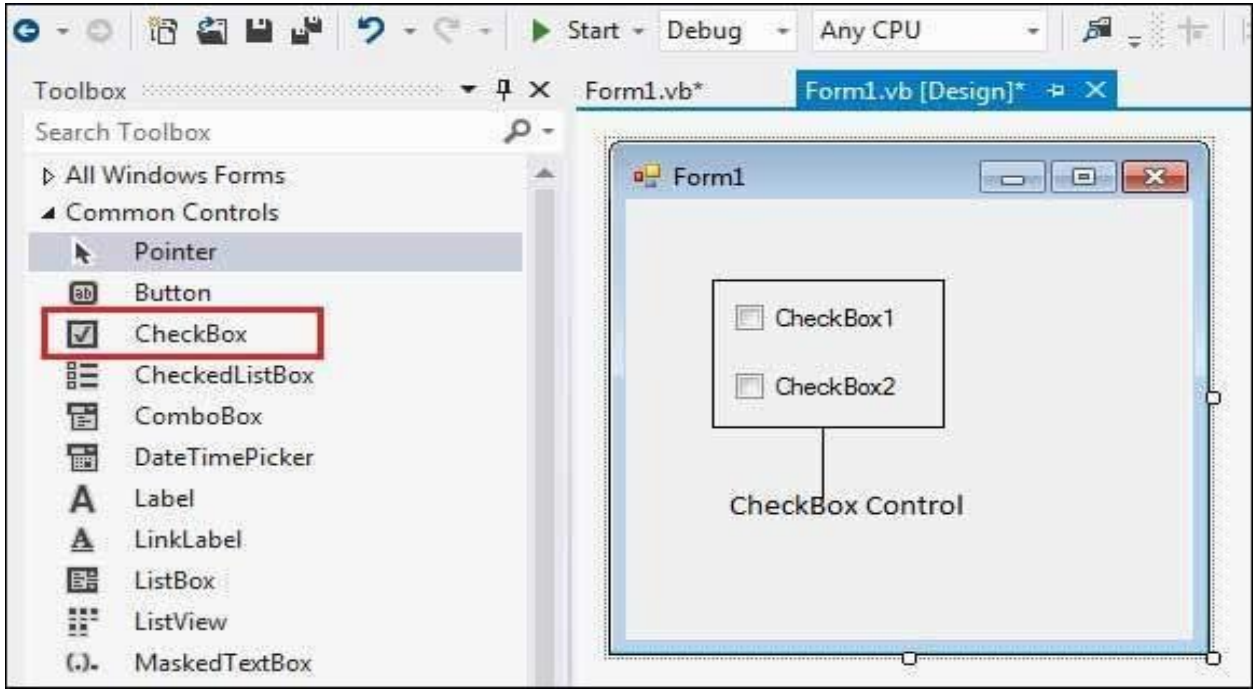
RadioButton Control

- This control is used to provide selection of only one option from the given group of option
- The **RadioButton** control can display text an image or both



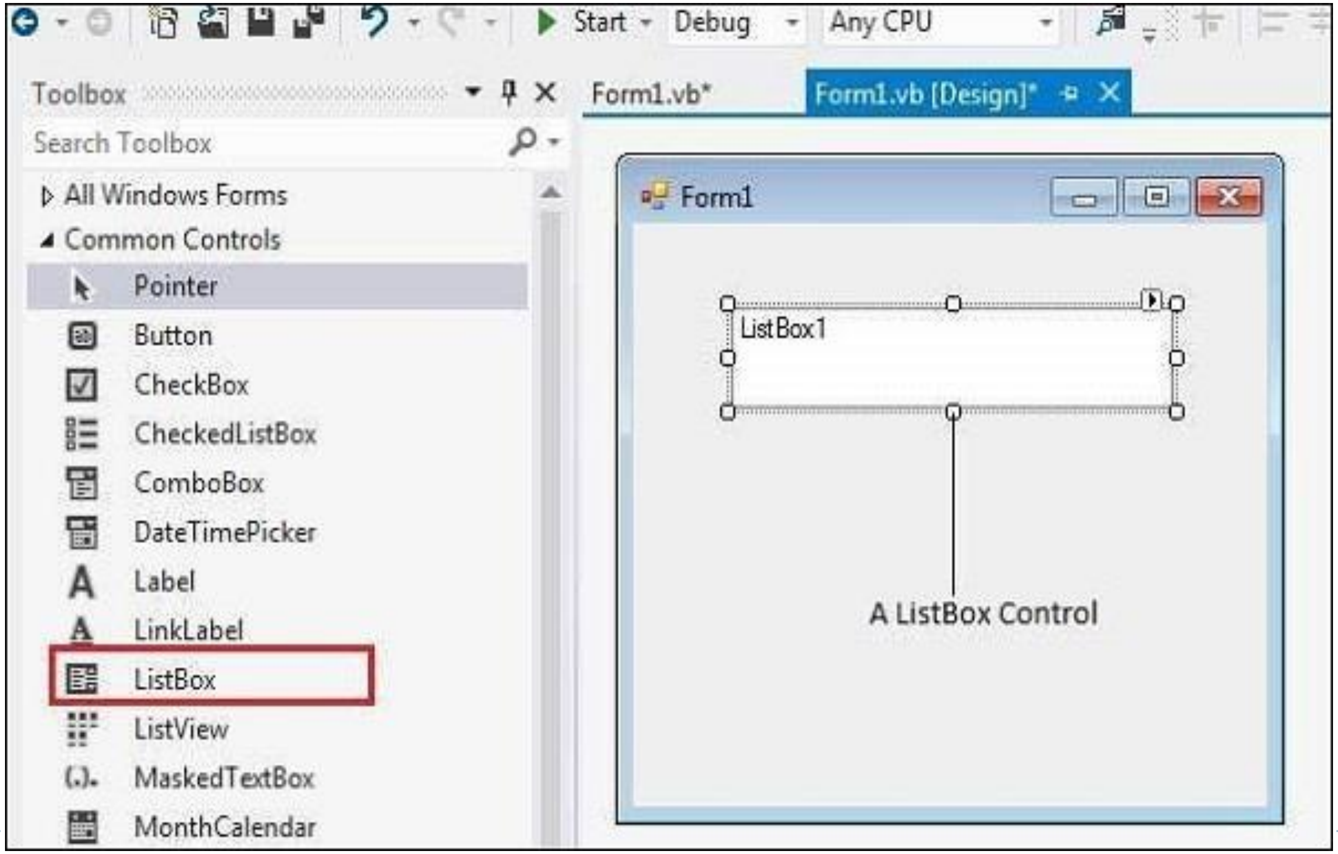
CheekBox Control

- This control is used to provide selection of more than one option from the given group of option
- **The** CheekBox control can display text, an image or both



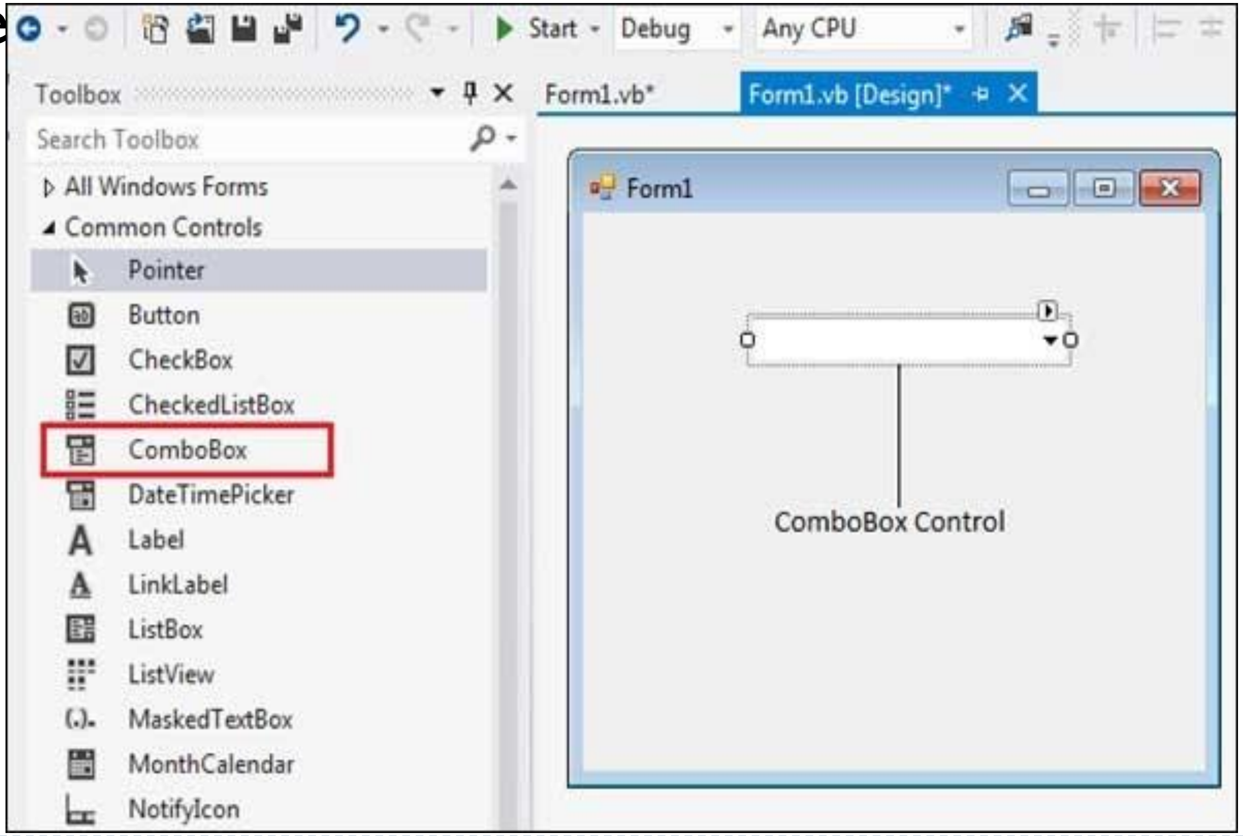
ListBox Control

- This control is used to display group of items to user so that users can select a single item or group of item randomly or range of items



comboBox Control

- Combo box is the combination of both textbox and listbox
- In combo box control, user can enter the data at run time and also we can display group of items to the use



Events

- Events are basically a user action like key press, clicks, mouse movements, etc., or some occurrence like system generated notifications. Applications need to respond to events when they occur.
- Clicking on a button, or entering some text in a text box, or clicking on a menu item, all are examples of events. An event is an action that calls a function or may cause another event. Event handlers are functions that tell how to respond to an event.
- VB.Net is an event-driven language. There are mainly two types of events −
- Mouse events
  - This event gets triggered when the pointer of the mouse enters the control.
-
- Keyboard events
  - These are the events that are triggered when the events are fired upon any action done on the keyboard. This includes actions such as keypress, keydown, enter, etc. Let us study some of the keyboard-based events in detail.

- To be continued in Lab……

EnD

Chapter 3:

Object-Oriented Fundamentals in VB.NET